MIL-F-50047 (CmlC)

29 June 1959
(See section 6)

MILITARY SPECIFICATION

FILTER, GAS, 600 CFM, C22

1. SCOPE

1.1 This specification covers a gas filter for use in removing toxic gases from the air forced or drawn through the filter.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-P-530 - Plywood, Flat Panel.
TT-E-527 - Enamel, Synthetic, Lustreless.
UU-P-271 - Paper, Wrapping, Waterproofed Kraft.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 4240

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DRAWINGS

CHEMICAL CORPS

C5-19-2060 - Filter, Gas, 600 CFM, C22 - Bill of Material.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing C5-19-2060 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the item shall conform to the specifications and drawings listed on Drawing C5-19-2060 and subsidiary drawings.

3.2 Manufacture and assembly. The filter shall be manufactured, assembled and marked as shown on Drawing D5-19-2034 and subsidiary drawings, which are listed on Drawing C5-19-2060.

3.3 Leakage. The filter shall not leak when tested as specified in 4.5.1.

3.4 Air resistance. The air resistance of the painted filter shall not exceed 1.25 inches of water when tested as specified in 4.5.2.

3.5 Workmanship. The filter shall be free from contamination (foreign matter), defects or damage that might impair its use.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Supplier's responsibility. Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

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4.2 Objective evidence. The supplier shall present objective evidence as required by the Government representative that all materials and components are in accordance with 3.1 and section 5 (see 6.3).

4.3 Alternate inspection (including testing) procedures. The supplier may utilize any alternate inspection procedures which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to its institution. In case of dispute, the procedures of this specification will govern.

4.4 Inspection provisions.

4.4.1 Lotting. A lot shall consist of the filters produced by one manufacturer from the same materials and under essentially the same manufacturing conditions. However, not more than two lots of C10 60 cubic feet per minute (c.f.m.), gas filters shall be represented in any one lot of C22 filters.

4.4.2 Sampling. Sampling for nondestructive inspection shall be conducted in accordance with Standard MIL-STD-105. The level and degree of inspection shall be specified in the contract.

4.4.3 Nondestructive examination. The regular production lot shall be inspected in accordance with the classification of defects and Standard MIL-STD-105.

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4.4.4 Classification of defects.

4.4.4.1 Filter, gas, 600 c.f.m., C22 - assembly. (Dwg. D5-19-2034)

Categories and defects		Acceptance standards
Critical		
*A.	Leakage	4.5.1
*B.	Air resistance	4.5.2
Major	AQL 1.0 percent defective	
1.	Components missing, incorrect, or incorrectly assembled	
2.	Damage affecting functioning	
3.	Adhesive material incorrectly applied	
4.	Marking incorrect	
Minor	AQL 2.5 percent defective	
51.	Contamination	
52.	Marking missing or illegible	

*Each filter in the lot shall be tested for leakage and air resistance.

4.4.4.2 Preparation for delivery. (section 5)

Categories and defects		
Major	AQL 4.0 percent defective	
1.	Inlet and outlet openings of the filter not covered as specified	
2.	Wood filler strips missing or inadequate to make a tight pack, where required.	
3.	Class and style of box not as specified.	
4.	Liner missing or incorrectly applied.	
5.	Closure of box incorrect.	
6.	Skids not as specified.	
7.	Box damaged.	
8.	Strapping loose or missing.	
9.	Marking incorrect, missing or illegible.	

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4.5 Tests.

4.5.1 Leakage. The filter shall be tested for leakage as follows:

(a) Connect the filter to a source of forced air as shown on figure 1.
(b) Regulate the airflow to the influent ducts of the filter to 600 c.f.m. by means of the orifice and draft gage.
(c) Spray isoamyl acetate (using a spray gun) into the inlet of the motor blower at a uniform rate of 1 ounce per minute for 1 minute.
(d) Wear a gas mask facepiece equipped with a hose and hold the open end of the hose against each effluent duct, in turn, and test the air by sniffing while moving the hose along the duct. Regulate the movement of the hose so that a thorough sampling of the effluent air will be obtained in the 1-minute period specified in (c) above. Detection of the odor of isoamyl acetate indicates leakage.
(e) If leakage is detected, repair leak and retest using above procedure. No more than three leakage retests of one filter shall be permitted.

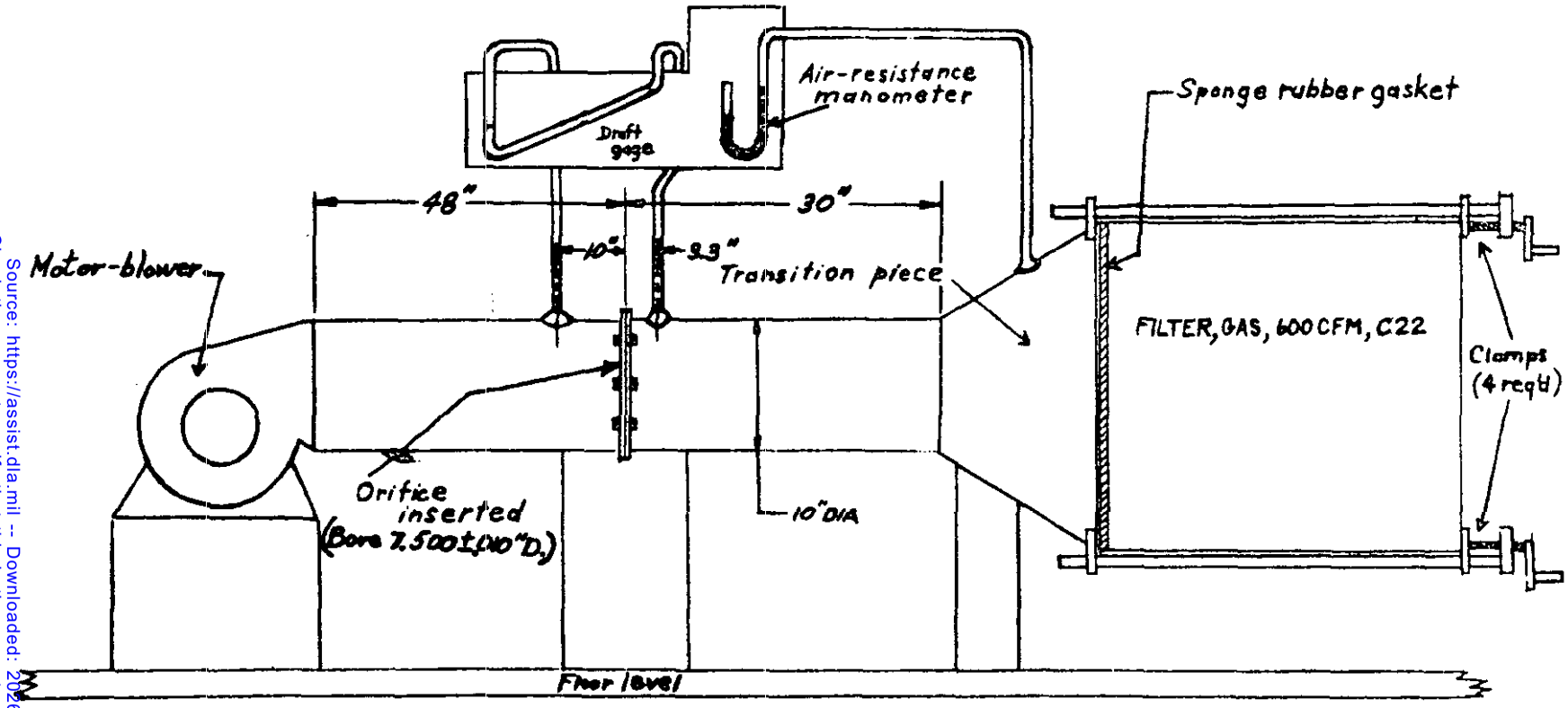
4.5.2 Air resistance. With the filter connected as shown on figure 1, the resistance shall be read on the manometer at an airflow of 600 c.f.m.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A. The inlet and outlet openings of the filter shall be covered with fitted 25-1/2 inch square by 1/4 inch thick exterior type, grade A-B, flat panel Douglas fir plywood, conforming to Specification NN-P-530, nailed in place. The plywood coverings shall be coated with synthetic lustreless enamel, color green, number 34087 of Specification TT-E-527. The filter shall then be packed in a class 2, style 2 or 2-1/2 nailed wood box conforming to Specification PPP-B-621. The interior sides, ends and top of the box shall be lined with type E-1 or E-2 waterproof barrier material conforming to Specification UU-P-271. To make a tight pack, wood filler strips shall be placed on the inside of the box between the sides, ends, top, and bottom of the box and the item. The box shall be closed in conformance with Specification PPP-B-621, with strapping to be in accordance with the appendix requirements thereto, using galvanized strapping. In addition, the box shall be modified by the addition of nominal 2 by 4 inch wood skids, conforming to

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NOTE:
Silver braze all taps flush with inside of duct.

Figure 1.- Test apparatus.

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the requirements for lumber as shown in the box specification, placed flatwise across the width of the box. The skids shall be positioned in such manner to provide for box stability and convenience of handling by fork lift trucks. The ends of the skids shall fit flush with the outside faces of the box and the lower end edges of the skids shall be beveled at 45° to the full depth of the skid.

5.1.2 Level B. Not applicable.

5.1.3 Level C. The filter shall be packed as specified in 5.1.1, except that a class 1, style 2 or 2-1/2 nailed wood box, conforming to Specification PPP-B-621, and coated-finish strapping shall be used.

5.2 Marking. In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. The filter covered by this specification is a component of Filter, Gas and Particulate, 600 CFM, M14 which is used for protecting fixed installations against toxic gases and aerosols.

6.2 Ordering data. Procurement documents should specify:

(a) The title, number and date of this specification.
(b) The level of packing required.

6.3 Objective evidence. Provisions for objective evidence and inspection records, and maintenance of inspection records will be specified by the procuring activity.

6.4 Pilot lot. A pilot lot may be required to evaluate the supplier's production and quality control systems. The contracting officer will specify whether a pilot lot is required and applicable pilot lot acceptance criteria.

6.5 Supersession data. This specification supersedes Chemical Corps Purchase Description 197-54-564A, dated 29 October 1958.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:

Army - Chemical Corps

Preparing activity:

Army - Chemical Corps

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